

David Picconi

Heinrich-Heine University
Düsseldorf

Place and date of birth:

Viterbo (Italy), 29 August 1988

Institute for Theoretical
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○ ResearchGate: [↗](#)

Education and research experience

Jun. 2024	Staff scientist <i>University of Düsseldorf, Germany</i>
–	Research group of Theoretical and Computational Chemistry (Prof. Shirin Faraji) <i>Permanent appointment expected in Spring 2026.</i>
Apr. 2022	Postdoctoral researcher <i>University of Groningen, the Netherlands</i>
May 2024	Research group of Theoretical Chemistry (Prof. Shirin Faraji)
Oct. 2019	Postdoctoral researcher <i>University of Potsdam, Germany</i>
Mar. 2022	Research groups of Theoretical Chemistry (Prof. Peter Saalfrank)
Sep. 2017	Postdoctoral researcher <i>Goethe-University Frankfurt am Main, Germany</i>
Sep. 2019	Research group of Theoretical Chemistry of Complex Systems (Prof. Irene Burghardt)
Oct. 2012	Doctoral Degree (PhD) <i>Technical University of Munich, Germany</i>
Jun. 2017	
thesis	PHOTODISSOCIATION DYNAMICS AND SPECTROSCOPY IN THE PRESENCE OF CONICAL INTERSECTIONS
supervisor	Prof. Wolfgang Domcke - <i>Technical University of Munich</i>
final grade	Highest distinction (<i>Summa cum laude</i>) Doctoral fellowship of the International Max Planck School for Advanced Photon Science .

2010–2012	Master Degree, Physical Chemistry, University of Pisa, Italy
thesis	DEVELOPMENT OF MODELS FOR QUANTUM DYNAMICAL SIMULATIONS OF PHOTO-EXCITED MOLECULES. APPLICATION TO THE $\pi\pi^*/n\pi^*$ INTERNAL CONVERSION IN THYMINE
supervisor	Dr. Fabrizio Santoro - <i>National Council of Research – Institute of Chemistry of Organometallics compounds (ICCOM-CNR)</i>
final grade	110/110 <i>cum laude</i>
2007–2012	Undergraduate course, Faculty of Sciences, Scuola Normale Superiore di Pisa, Italy Initiative for excellent education, upon competitive selection ↗

Research interests

- Development of theories, codes and algorithms for high-dimensional molecular quantum dynamics and mixed quantum-classical simulations of chemical systems. Development of methods to describe multi-mode multi-state open quantum systems, analyse decoherence and dissipation.
- Simulation of photo-initiated processes in molecules and materials in the time and energy domain: Internal conversion and intersystem crossing, nonadiabatic phenomena, photodissociation, photoisomerisation, charge/energy transfer, exciton dissociation and annihilation, photoinduced chemical reactions, etc.
- Quantum chemical description of molecular potential energy surfaces. Modelling of ground and excited states, as well as multi-state reaction pathways.
- First-principles simulation of optical spectroscopies of gas- and condensed-phase molecular systems: One-photon processes (absorption, emission, circular dichroism), resonance Raman scattering, two-photon absorption, pump-probe, four-wave mixing and photoelectron signals for different ranges of wavelengths (infrared, ultraviolet, X-ray). Development of theoretical models to correlate the spectroscopic signals to the underlying molecular dynamics.

Active and past collaborations

- Prof. Markus **Gühr**, German Electron Synchrotron DESY and University of Hamburg, Germany.
Ultrafast photochemistry investigated using time-resolved X-ray spectroscopy. (2019–present)
- Prof. Irene **Burghardt**, University of Frankfurt, Germany.
Theoretical methods for high-dimensional quantum dynamics and open quantum systems. Multi-pulse vibronic spectroscopy. (2019–present)

- Prof. David **McCamant**, University of Rochester, United States.
Resonance Raman and time-resolved spectroscopy of excitonic dimers. (2025–present)
- Prof. Jens **Bredenbeck**, University of Frankfurt, Germany.
Mixed vibrational-electronic spectroscopy (infrared-ultraviolet). (2022)
- Prof. Jeffrey A. **Cina**, University of Oregon, United States.
Theoretical description of multi-dimensional optical spectroscopy. (2017–2019)

Bibliometric data

Source: SCOPUS database

Citations 441

H-index 12

Author position information for the last 20 publications in the period 2015–2024

Position	%	No. of publications	Average citations
First author	45%	9	10
Last author	10%	2	6
Co-author	30%	6	8
Single author	15%	3	10
Corresponding author	70%	14	13

Invited and contributed talks at international conferences

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|-------------|---|
| contributed | NOVEL APPROACHES TO SIMULATE THE DYNAMICS OF MULTI-MODE NONADIABATIC QUANTUM SYSTEMS COUPLED TO A FLUCTUATING ENVIRONMENT, at the 16th Femtochemistry Conference (FEMTO 16) ↗ , Trieste, Italy, June 2025 |
| contributed | NOVEL APPROACHES TO SIMULATE THE DYNAMICS OF MULTI-MODE NONADIABATIC QUANTUM SYSTEMS COUPLED TO A FLUCTUATING ENVIRONMENT, at "Quantum Dynamics and Spectroscopy of Functional Molecular Materials and Biological Photosystems" ↗ , Les Houches, France, March 2025 |
| invited | HIGH-DIMENSIONAL QUANTUM DYNAMICS WITH DISSIPATION AND DECOHERENCE: CHALLENGES AND APPLICATIONS, at "High Dimensional Quantum Dynamics: Challenges and Applications (HDQD2024)" ↗ , Hamburg, Germany, July 2024 |
| contributed | PHOTOPHYSICS, PHOTOCHEMISTRY AND SPECTROSCOPY USING HIGH-DIMENSIONAL QUANTUM WAVE PACKETS, at "NWO Physics", Veldhoven, the Netherlands, January 2024 |

invited	PHOTOPHYSICS, PHOTOCHEMISTRY AND SPECTROSCOPY USING HIGH-DIMENSIONAL QUANTUM WAVE PACKETS, at "Quantum Dynamics and Spectroscopy of Functional Molecular Materials and Biological Photosystems" ↗ , Les Houches, France, September 2023
contributed	PHOTOPHYSICS, PHOTOCHEMISTRY AND SPECTROSCOPY USING HIGH-DIMENSIONAL QUANTUM WAVE PACKETS AND <i>ab initio</i> POTENTIAL ENERGY SURFACES, at the International Conference on Photochemistry (ICP2023) ↗ , Sapporo, Japan, July 2023
contributed	ULTRAFAST MOLECULAR PHOTODYNAMICS AND THEIR SPECTROSCOPIC DETECTION, at the 28th IUPAC Symposium on Photochemistry (PhotoIUPAC) ↗ , Amsterdam, the Netherlands, July 2022
contributed	ULTRAFAST MOLECULAR PHOTODYNAMICS AND THEIR SPECTROSCOPIC DETECTION, at "High-Dimensional Quantum Dynamics: Challenges and Applications" (HDQD2022) ↗ , Groningen, the Netherlands, July 2022
contributed	MOLECULAR PHOTODYNAMICS AND ITS MANIFESTATION IN UV-VIS AND X-RAY TIME-RESOLVED SPECTRA: INSIGHTS FROM QUANTUM CHEMISTRY AND QUANTUM DYNAMICS, at the 57th Symposium on Theoretical Chemistry ↗ , Würzburg, Germany, September 2021
contributed	MOLECULAR PHOTODYNAMICS AND ITS MANIFESTATION IN UV-VIS AND X-RAY TIME-RESOLVED SPECTRA: INSIGHTS FROM QUANTUM CHEMISTRY AND QUANTUM DYNAMICS, at the International Conference on Photochemistry ↗ , Geneva, Switzerland, July 2021
invited	MOLECULAR PHOTODYNAMICS AND ITS MANIFESTATION IN UV-VIS AND X-RAY TIME-RESOLVED SPECTRA: INSIGHTS FROM QUANTUM CHEMISTRY AND QUANTUM DYNAMICS, at "Light Induced Processes in Physical, Chemical, and Biological Systems" ↗ , Zagreb, Croatia, June 2021
invited	MOLECULAR PHOTODYNAMICS AND ITS MANIFESTATION IN UV-VIS AND X-RAY TIME-RESOLVED SPECTRA: INSIGHTS FROM QUANTUM CHEMISTRY AND QUANTUM DYNAMICS, at "Quantum Dynamics and Spectroscopy of Functional Molecular Materials and Biological Photosystems" ↗ , Les Houches, France, May 2021
contributed	TIME-RESOLVED SPECTRA OF I ₂ IN A KRYPTON CRYSTAL BY G-MCTDH SIMULATIONS: NONADIABATIC DYNAMICS, DISSIPATION AND ENVIRONMENT DRIVEN DECOHERENCE, at "Beating the complexity of matter through the selectivity of X-rays. Dynamic pathways in multidimensional landscapes" ↗ , Berlin, Germany, September 2019
contributed	TIME-RESOLVED SPECTRA OF I ₂ IN A KRYPTON CRYSTAL BY G-MCTDH SIMULATIONS: NONADIABATIC DYNAMICS, DISSIPATION AND ENVIRONMENT DRIVEN DECOHERENCE, at "Quantum effects in complex systems: Faraday Discussion" ↗ , Coventry, United Kingdom, September 2019

invited	QUANTUM DYNAMICS AND SPECTROSCOPY USING G-MCTDH WAVEFUNCTIONS. NONADIABATIC PHOTOPHYSICS AND VIBRATIONAL COHERENCES OF I ₂ IN SOLID Kr, at “Quantum dynamics and spectroscopy in condensed-phase materials and bio-systems” ☞ , Telluride, United States, June 2019
contributed	PHOTODYNAMICS AND SPECTROSCOPY OF HALOGENS EMBEDDED IN RARE GAS CRYSTALS. QUANTUM DYNAMICS OF I ₂ IN SOLID Kr, at the “54th Symposium on Theoretical Chemistry” ☞ , Halle (Saale), Germany, September 2018.
contributed	PHOTODISSOCIATION REACTIONS THROUGH CONICAL INTERSECTIONS: INSIGHTS INTO THE ABSORPTION SPECTRUM AND THE PRODUCT STATE DISTRIBUTIONS, at “High dimensional quantum dynamics: challenges and opportunities” ☞ , Rostock, Germany, September 2016.
invited	INSIGHTS INTO THE ABSORPTION SPECTRUM AND THE PRODUCT STATE DISTRIBUTIONS OF PHOTODISSOCIATING MOLECULES, at “Non-Equilibrium Statistical Physics: from molecular materials to theoretical engineering. Honoring Professor Vladimir Chernyak 60th Birthday” ☞ , Telluride, United States, July 2016
contributed	TRACKING REACTIVE CONICAL INTERSECTIONS IN PHOTODISSOCIATION VIA INTERNAL STATE DISTRIBUTIONS OF (INTERACTING) FRAGMENTS, at the 13th International Workshop on Quantum Reactive Scattering ☞ , Salamanca, July 2015
contributed	TRACKING REACTIVE CONICAL INTERSECTIONS IN PHOTODISSOCIATION VIA INTERNAL STATE DISTRIBUTIONS OF (INTERACTING) FRAGMENTS, at the 5th Summit Meeting “Excitation energy transfer processes in physical, chemical and biological systems”, Munich, Germany, July 2015
contributed	A STRATEGY FOR NONADIABATIC QUANTUM-CLASSICAL DYNAMICS OF SEMIRIGID MOLECULES. INVESTIGATING THE PHOTOSTABILITY OF NUCLEOBASES, at the Workshop of the Section of Theoretical and Computational Chemistry of the Italian Chemical Society, Padova, Italy, February 2013
contributed	MODELS FOR QUANTUM DYNAMICAL SIMULATIONS OF PHOTOEXCITED MOLECULES. INVESTIGATING THE PHOTOSTABILITY OF NUCLEOBASES, at the International Meeting on Atomic and Molecular Physics and Chemistry ☞ , Pisa, Italy, September 2012

Additionally, I have given > 10 university colloquia and invited seminars in different research groups.

Additional scientific activities

Participation in funded projects after the doctorate:

- "Watching chemistry happen with light", VIDI Grant funded by the *Nederlandse Organisatie voor Wetenschappelijk Onderzoek (NWO)* (PI: Prof. S. Faraji) [Publication 24]
- "Gaussian methods for quantum dynamics of complex systems", funded by the *German Israeli Foundation (GIF)* (PI: Prof. I. Burghardt, 2017-2109) [Publication 15]
- "Probing thionucleobase dynamics at the sulfur L-edge", funded by the *Deutsches Forschungsgemeinschaft (DFG)* (PIs: Prof. P. Saalfrank and Prof. M. Gühr, project n. 445713302, started in 2020) [Publications 21, 20 and 18]
- "Theoretical description of a novel dye-sensitized solar cell", funded by the *Deutsches Forschungsgemeinschaft (DFG)* (PI: Prof. P. Saalfrank, project n. 251801169, started in 2014) [Publications 22 and 19]

Reviewing activity

Reviewer for the following journals:

- *Nature Chemistry*
- *Nature Communications*
- *The Journal of Physical Chemistry Letters*
- *Journal of Chemical Theory and Computation*
- *The Journal of Chemical Physics*
- *Physical Chemistry Chemical Physics*
- *Journal of Molecular Liquids*

For the years 2021–2025, 20 reviews can be verified in my [ORCID page](#)

Grant evaluation

- (2025) Assessor for a Research Proposal of the German Science Agency (DFG)
- (2022–2023) Assessor for the XS Grant funded by Dutch Science Agency (NWO) for three calls (10 proposals evaluated for each call).

Conference organization

- Since 2024 I am part of the scientific panel organizing the biannual conference and school *Quantum dynamics and spectroscopy of functional molecular materials and biological photosystems* [↗](#) (Chair of the board: Dr. Jérémie Leonard, Univ. Strasbourg). The next edition will be held in Les Houches, France, in March 2027.
- 2025. Scientific organization of the CECAM workshop *Future directions in non-adiabatic dynamics: towards complex systems and long time scales* in Zaragoza, Spain [↗](#)
- 2022. Part of the local organization committee of the conference *High-Dimensional Quantum Dynamics* in Groningen, the Netherlands (chair: Prof. Shirin Faraji)

Conference chairing

I have been assigned as chairman for one session at the following workshops:

- *Quantum dynamics and spectroscopy of functional molecular materials and biological photosystems* [↗](#), Les Houches, September 2023 and March 2025
- IUPAC|CHAINS, The Hague, August 2023
- *High-Dimensional Quantum Dynamics*, Groningen, July 2022
- *Quantum dynamics and spectroscopy of functional molecular materials and biological photosystems* [↗](#), Les Houches, May 2021
- *Quantum dynamics and spectroscopy in condensed-phase materials and bio-systems* [↗](#), Telluride, June 2019
- 5th Summit Meeting “Excitation energy transfer processes in physical, chemical and biological systems”, Munich, July 2015

Popular press release

- *Watching the charge move in photoexcited molecules*, press release of the Deutsches Elektronen-Synchrotron DESY [↗](#)

Fellowships and awards

2021 I co-wrote a proposal for beamtime allocation at the Elettra Synchrotron in Trieste (spokesperson for the collaboration: Prof. M. Gühr, University of Potsdam). The proposal was **accepted**, and time-resolved X-ray spectroscopic experiments were performed in early 2023, and are presently being analyzed.
I am coordinating the theory part of the project.

2012–2017 Fellowship of the [International Max Planck Research School for Advanced Photon Science](#) (assigned by an open competition)

July 2015 *Aaron Kuppermann QRS Best Poster Prize* awarded by the *Journal of Physical Chemistry A*, at the [13th International Workshop on Quantum Reactive Scattering](#)

SNS undergraduate fellowship Students admitted at the Excellence Initiative of the [Scuola Normale Superiore \(SNS\)](#) attend the ordinary courses at the University of Pisa, but are required to complement them with additional classes taught by professors of the SNS. In their exams SNS students are required to score average marks of at least 27/30 yearly and no mark below 24/30 in order to maintain the scholarship.

Teaching experience

For each course, the course coordinator is given in parenthesis, the level of the students (bachelor/master/PhD) is reported, and my role is explained briefly

At the Heinrich-Heine University Düsseldorf

- 2025-2026 *Quantum and semiclassical molecular dynamics* (Dr. David Picconi, Prof. Shirin Faraji)
Master in Chemistry
I am co-coordinator of this course, I give the lectures and the tutorials, I conduct the examinations and grade.
- 2024-2025 *Computer Praktikum for the course of Molecular Photonics and Excited-State Processes* (Prof. Shirin Faraji)
Master in Chemistry
I have been a co-lecturer and teaching assistant for the computational exercises.
- 2024-2025 *Exercises of Quantum and Computational Chemistry (QCCC)* (Prof. Shirin Faraji, Dr. David Picconi)
Bachelor in Chemistry
I coordinated the organization of the exercises for the course and gave tutorial lectures.

At the University of Groningen

- 2023 *Topics in Chemistry with Python* (Prof. Shirin Faraji)
Master in Chemistry, Theoretical and Computational Chemistry Master (TCCM)
I have been a teaching assistant and I have given part of the lectures.
- 2023 *Molecular Quantum Mechanics II* (Prof. Shirin Faraji)
Master in Chemistry, Theoretical and Computational Chemistry Master (TCCM)
I have given part of the lectures.

At the University of Potsdam

- 2020-2022 *Theoretical Photochemistry and Spectroscopy* (Dr. David Picconi)
Master & PhD in Chemistry or Physics
I fully coordinated this course for two semesters, I gave the lectures and the tutorials, I conducted the examinations and graded.

At the Goethe-University of Frankfurt

- 2017-2019 *Theoretical Photochemistry* (Prof. Irene Burghardt)
Master in Chemistry
I gave the tutorial lectures (in active participatory or peer-review format) and some frontal lessons. I conducted the examinations and graded.
- 2018 *Theoretical Chemistry II* (Prof. Irene Burghardt)
Bachelor & Master in Chemistry or Biophysics
I gave some tutorial lectures.

At the Technical University of Munich

- 2016 *Laboratory of Programming and Numerical Methods* (Dr. C. Scheurer)
Master in Chemistry
I supervised the students in the computer practicum ($\approx 75\%$ of the course time)
- 2015 *Quantum Dynamics and Spectroscopy* (Prof. W. Domcke)
Master & PhD in Chemistry or Physics
Tutorial lectures.
- 2014 *Chemistry for Physics* (Priv.-Doz. S. Yu. Grebenshchikov)
Bachelor in Physics
Tutorial lectures
- 2013-2015 *Quantum Mechanics* (Priv.-Doz. S. Yu. Grebenshchikov)
Bachelor in Chemistry
Tutorial lectures and training of the teaching assistants
- 2013 *Analytical Chemistry* (Dr. S. Breitenlechner and Dr. A. Bauer)
Bachelor in Management and Technology (TUM-BWL)
Supervision during laboratory practicum and grading

Additional teaching activities

- 2023-2025 Lecturer at two editions of the School "Quantum Dynamics and Spectroscopy of Functional Molecular Materials and Biological Photosystems", Les Houches (August 2023 [↗](#) and February 2025 [↗](#))

Supervision of students

- 2025 Co-supervisor of a doctoral student (Jan Ortlepp) in the Group of Theoretical Chemistry at the University of Düsseldorf (Prof. S. Faraji)
- 2025 Supervisor of a master student (Jan Ortlepp) in the Group of Theoretical Chemistry at the University of Düsseldorf (Prof. S. Faraji)
- 2024–
present Daily supervisor of a doctoral student (Kors Doedens) in the Group of Theoretical Chemistry at the University of Düsseldorf (Prof. S. Faraji)
- 2022 Daily supervisor of a doctoral student (Edison X. Salazar) in the final part of his PhD in the Group of Theoretical Chemistry at the University of Groningen (Prof. S. Faraji)

2020– present	Co-supervision of a doctoral student (Maximiliane Horz) in one of her PhD sub-projects in the Group of Theoretical Chemistry at the University of Frankfurt (Prof. Irene Burghardt)
2017	Daily supervision of a master student (Johannes Ehrmaier) in the group of Prof. W. Domcke, at the Technical University of Munich. The work has resulted in a publication.
2014	Daily supervision of a bachelor student (Carlos Mejuto Zaera) in the group of Prof. W. Domcke, at the Technical University of Munich

Research technical skills

- Programming in FORTRAN and PYTHON
- Linux operating system and bash scripting
- Q-CHEM, GAUSSIAN, ORCA, TURBOMOLE, MOLPRO, and OPENMOLCAS packages for quantum chemistry
- Data analysis and machine learning libraries: `scikit-learn`, `pandas`, `tensorflow`, `keras`
- Techniques for quantum wave packet propagation. Autonomous development of codes for quantum dynamics based on the multi-configurational time-dependent Hartree (MCTDH) method, the variational Gaussian wave packets (vMCG), and their combination (G-MCTDH).
- MATHEMATICA and ORIGIN packages

Language skills

- Italian, native
- English, fluent
- German, B2 level
- Spanish, fluent